

Yash Jain

+1 604-374-3265 | y29jain@uwaterloo.ca | github.com/yashjain128 | linkedin.com/in/yashjain128

EDUCATION

University of Waterloo

Waterloo, ON

Candidate for Bachelor of Software Engineering, CGPA: 3.9/4.0

- USA Computing Olympiad Gold, Engineering Ambassador

EXPERIENCE

Software Developer

Sep 2025 – Present

Waterloo Rocketry

Waterloo, ON

- Developed a real-time network bridge to transmit live rocket telemetry to ground control dashboards, integrating a type-safe serialization protocol using Pydantic to validate 30+ sensor fields and eliminate malformed packets.
- Created a pre-competition confidence check using Pytest, achieving high code coverage for message parsing, which was also used to identify three deprecated protocols in the codebase

Software Research Assistant

Jun 2023 – Aug 2024

Space and Atmospheric Instrumentation Lab at Embry Riddle Aeronautical University

Daytona, FL - Remote

- Built a telemetry engine for NASA-funded mission rockets, using non-blocking sockets to handle live asynchronous UDP streams, preventing UI lockups and data loss during network drops
- Engineered a noise-resilient sensor parsing pipeline by implementing NumPy cross-correlation to dynamically identify sync frames and recover fragmented packets from raw byte streams
- Decoded hardware signals in real-time by vectorizing bitwise extractions, masking, and CRC-16 validations into usable housekeeping and geodetic data
- Increased rendering performance by 4x over a legacy MATLAB system by implementing an OpenGL-backed visualization layer (VisPy) for continuous time-series plotting

Team Captain and Programmer

Sep 2022 – Jun 2025

Salish Robotics Team

Vancouver, BC

- Trained autonomous navigation routines in C++ by implementing PID controllers for precision steering and Pure Pursuit for smooth path-tracking, optimizing for consistency during competition
- Managed the 3D design lifecycle in OnShape to coordinate a team of 8 by defining clear mechanical interfaces that allowed hardware and software development to proceed in parallel
- Secured the Judges and Create awards by implementing an iterative testing protocol to analyze and resolve mechanical failure points

PROJECTS

A-List Housing | *Python, TypeScript, React, Flask, Selenium*

- Deployed a full-stack platform to provide data-driven property evaluations and macroeconomic forecasts
- Created an engine to forecast macro-economic housing trajectories by implementing an XGBoost hedonic pricing classifier trained on 100k+ property listings
- Developed a k-nearest neighbours recommendation feature to identify comparable property sales, utilizing historical market data gathered through Selenium web scraping

Wunder Challenge | *PyTorch, ONNX, Pandas*

- Trained a PyTorch GRU model to forecast high-frequency time-series data by analyzing rolling windows of historical events
- Sped up live predictions by exporting the model to ONNX Runtime and writing a custom script to manage sequence padding and rolling memory histories.

TECHNICAL SKILLS

Languages: C, C++, Python, MATLAB, Verilog, SQL, Bash, Javascript, HTML/CSS

Technologies: NumPy, PyTorch, Pandas, Flask, PyQt, VisPy, Matplotlib, PyTest, Selenium, React, Node.js

Developer Tools: Git, Cloudflare, Github Pages, Firebase, Linux systems